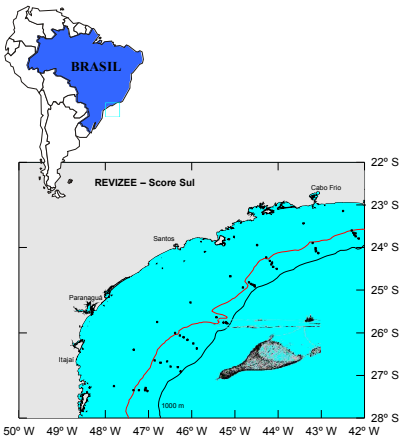
FEEDING IN THE DEEP SEA: DIET OF THREE DEMERSAL FISHES OF THE SOUTHERN BRAZIL

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INTRODUTION

Studies on diet can provide meaningful information about the biology of a species, and are fundamental for understanding the trophic relationships in an ecosystem. We investigated the feeding ecology of three abundant demersal fish species, *Antigonia capros, Ariomma bondi* and *Ventrifossa macropogon,* in the Brazilian Exclusive Economic Zone.



Map indicating the sampling sites. Red line = estimate shelf break localization. In the right side a balloon trawl draw exemplifying the survey trawl.

Antigonia capros



Bernardes *et al.* 2005

Ariomma bondi



Bernardes *et al.* 2005

Ventrifossa macropogon



Bernardes *et al.* 2005

MATERIAL AND METHODS

Sampling was conducted in 2001 and 2002, during austral winter/spring and summer/fall, off the southeastern Brazilian coast between Cabo de Santa Marta Grande (State of Santa Catarina) and Cabo Frio (State of Rio de Janeiro), during the REVIZEE Score-Sul program. Samples were taken by bottom-trawling, at depths between 100 and 600 m.



Pictures illustrating the work steps, bottom trawling, sorting fishing, stomach removal, food items identification.

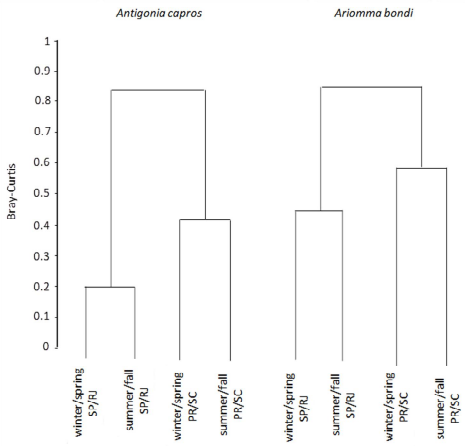
Each specimen stomach was removed, and prey items were identified and weighed. A feeding index was calculated to evaluate the importance of prey items, according to areas, seasons, size classes and depths, and overlap in diet composition was assessed through cluster analysis of this index. Also, the feeding activity was assessed for each species through the degrees of stomach repletion (fullness) and food digestion.

RESULTS AND DISCUSSION

All three species fed mainly on **crustaceans**, and the main factor affecting diet composition was the sampling area; season and time of day had secondary effects. It can be seen that invertebrates, mainly the demersal and benthic ones, are important to maintain the population of the fish species studied.

Table of food index (FI%) for each species related to area, season, depth and individuals size

		Antigonia capros					*Ariomma bondi*					*Ventrifossa macropogon*			
		Gastropoda	Cephalopoda	Polychaeta	Crustacea	Teleostei	Gastropoda	Cephalopoda	Crustacea	Echinodermata	Teleostei	Cephalopoda	Polychaeta	Crustacea	Teleostei
winter/spring	SP/RJ	<1	<1	<1	99	<1	-	-	100	-	<1	<1	<1	94	6
	PR/SC	<1	2	<1	98	-	100	-	-	-	<1	-	-	-	-
summer/fall	SP/RJ	<1	1	<1	99	<1	-	<1	98		2	-	-	93	7
	PR/SC	<1	3	-	97	-	14	-	86	<1	-	-	-	100	-
Depth	100m	<1	<1	1	99	<1	-	-	-	-	-	-	-	-	-
	150m	<1	<1	<1	99	<1	1	-	99	-	<1	-	-	-	-
	200m	<1	1	<1	99	1	8	-	93	-	-	-	-	-	-
	300m	<1	11	<1	89	<1	-	-	92	-	8	-	-	-	-
	400m	-	-	-	-	-	32	3	57	1	9	-	-	-	-
	500m	1	4	<1	95	<1	-	-	100	-	-	-	-	-	-
	600m	-	-	-	-	-	-	-	-	-	-	<1	<1	95	5
Size classes	40-50	-	2	-	98	-	-	-	-	-	-	-	-	-	-
	50-60	1	1	1	97	1	-	-	-	-	-	-	-	-	-
	60-70	-	1	<1	99	-	-	-	-	-	-	-	-	-	-
	70-80	-	<1	-	100	-	-	-	-	-	-	-	-	-	-
	80-90	<1	<1	<1	99	-	-	-	-	-	-	-	-	-	-
	90-100	-	<1	<1	100	<1	-	-	-	-	-	-	-	-	-
	100-110	<1	1	<1	99	<1	44	-	56	-	-	-	-	-	-
	110-120	<1	<1	<1	100	-	-	-	100	-	-	-	-	-	-
	120-130	<1	1	<1	99	<1	9	-	91	-	-	-	-	-	-
	130-140	<1	1	<1	99	<1	-	-	100	-	<1	-	-	-	-
	140-150	<1	1	<1	99	1	-	-	100	-	-	-	-	-	-
	150-160	<1	<1	<1	99	1	-	-	100	-	-	-	-	-	-
	160-170	-	-	-	100	-	-	1	87	<1	11	-	-	-	-
	170-180	-	-	-	-	-	23	-	77	-	-	-	-	-	-
	180-190	-	-	-	-	-	16	-	81	-	2	-	-	-	-
	160-180	-	-	-	-	-	-	-	-	-	-	-	-	100	-
	180-200	-	-	-	-	-	-	-	-	-	-	-	<1	100	-
	200-220	-	-	-	-	-	-	-	-	-	-	-	-	100	-
	220-240	-	-	-	-	-	-	-	-	-	-	-	-	100	-
	240-260	-	-	-	-	-	-	-	-	-	-	-	<1	100	<1
	260-280	-	-	-	-	-	-	-	-	-	-	<1	<1	100	-
	280-300	-	-	-	-	-	-	-	-	-	-	-	-	80	20
	300-320	-	-	-	-	-	-	-	-	-	-	-	-	98	2
	>320	-	-	-	-	-	-	-	-	-	-	-	-	18	82



Similarities of diet composition for *Antigonia Capros* and *Ariomma bondi* between sampled areas and seasons

Changes in the local availability of invertebrates imply in changes in the main resource consumed by these species.

Table of repletion index (RI%); Repletion degree (RD) when RD0 means empty stomach and RD3 is stomach completely full; and digestion degree (DD) when DD1 means fresh food and DD4 means completed digested food, for each species.

		Antigonia capros								*Ariomma bondi*								*Ventrifossa macropogon*			
		winter/spring				summer/fall				winter/spring				summer/fall							
		06:00	09:00	12:00	15:00	06:00	09:00	12:00	15:00	06:00	09:00	12:00	15:00	06:00	09:00	12:00	15:00	06:00	09:00	12:00	15:00
Repletion index	RI%	0.32	0.39	0.26	0.49	0.33	0.30	0.65	0.6	0.54	0.18	0.01	0.14	0.44	0.61	0.40	0.38	0.54	0.18	-	0.14
Repletion degree	RD0	6	33	36	8	-	-	-	-	12	62	93	76	30	23	65	81	9	-	-	-
	RD1	25	-	60	9	48	41	20	17	54	37	0	21	50	47	29	13	68	50	-	40
	RD2	62	33	5	68	52	59	70	83	35	1	7	2	16	27	5	3	21	50	-	60
	RD3	7	33	-	16	-	-	11	-	-	-	-	-	4	3	-	3	2	-	-	-
Digestion degree	DD1	5	25	-	6	-	-	14	83	-	-	-	-	-	-	-	-	7	-	-	-
	DD2	16	25	4	43	10	4	11	17	30	-	-	-	1	2	-	-	5	50	-	-
	DD3	79	50	96	51	90	96	75	-	48	21	100	50	22	22	15	35	88	50	-	100
	DD4	-	-	-	-	-	-	-	-	22	79	-	50	76	76	85	65				

CONCLUSION

The plasticity in the food consumption of three species can confer a high resilience to their population. On the other hand, due to their high dependence to the benthic and demersal invertebrate community, activities that systematically remove large amount of these organisms from the ecosystem, as it is often the case of bottom trawling, can give a large negative impact on three species

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